

ME1103

Principles of Mechanics and Materials

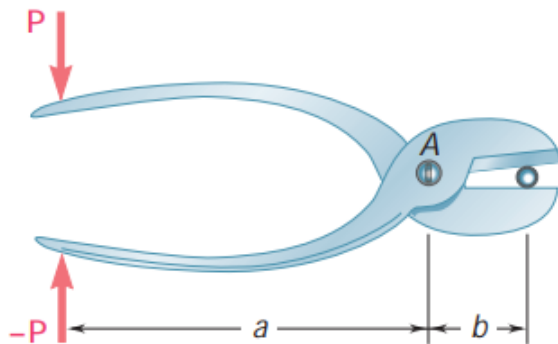
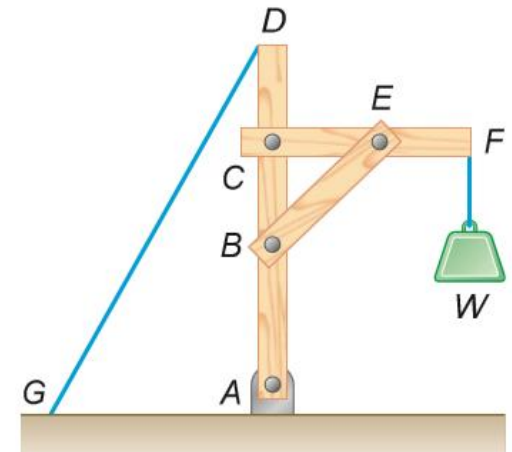
ANALYSIS OF FRAME/MACHINE

Learning Outcomes

- Understand the characteristics of frame and machine as pin joined assembly with multiple-force members.
- Able to perform static analysis on frame and machine with FBDs of the structure and/or its components and sub-systems.
- Understand the concept of dry friction and its relation to the normal contact force and condition of the contact interface (in terms of frictional coefficient).

Frames and Machines

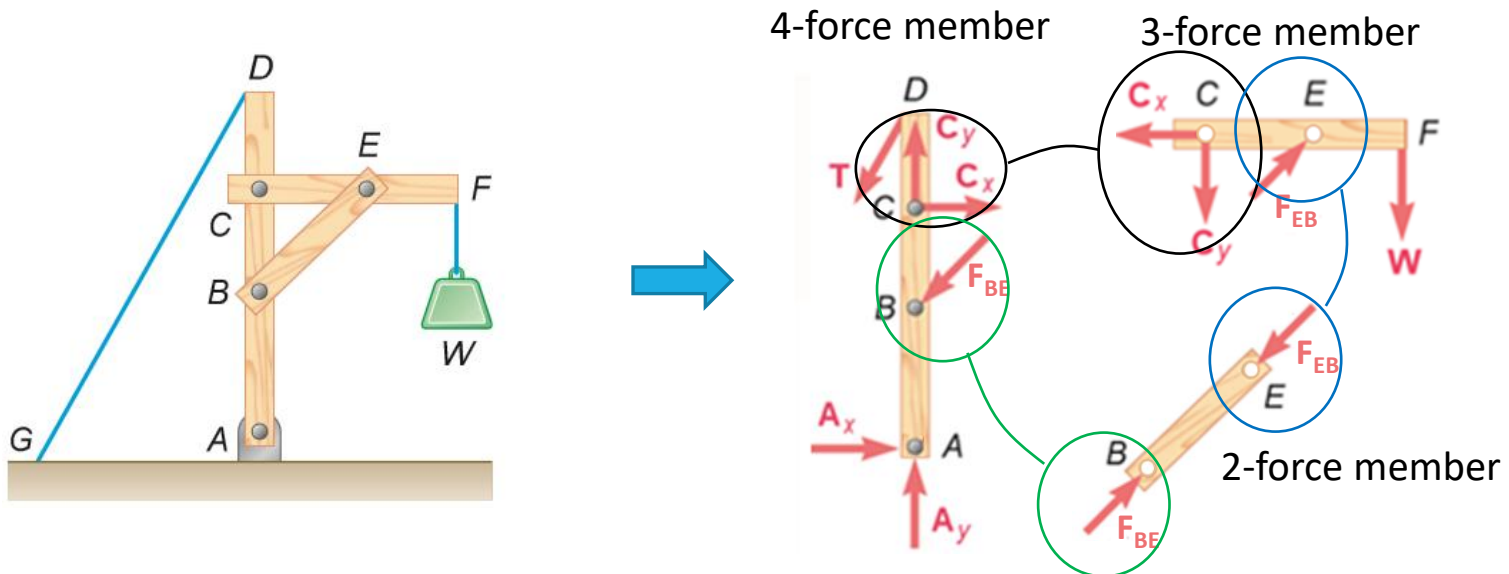
- Frames and machines are structures connected by pin joints that consist of at least one multi-force members, i.e. members that are subjected to more than two forces that are applied at different locations.
- Frames are usually rigid structures that are designed to support loads. That is, the structure will not collapse when they are isolated.



- Machines contains moving parts and are designed to transmit and/or modify the effect of forces.

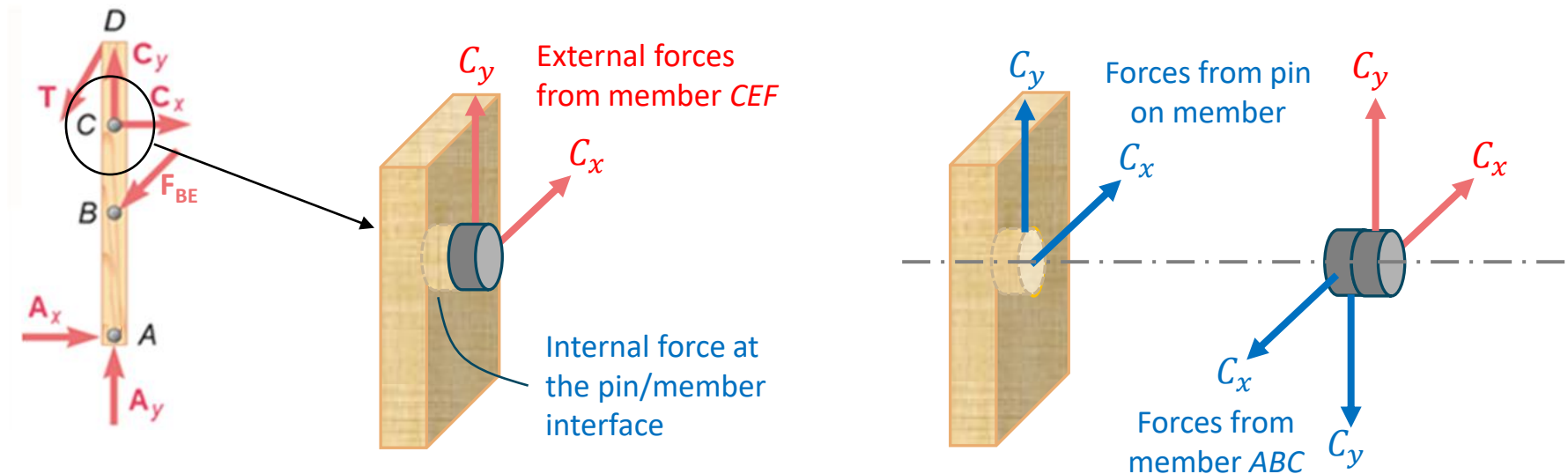
Analysis of Frames and Machines

- Consider the crane example. Here, members $ABCD$ and CEF are multi-force members, while BE is a two-forces member.
- For multi-force members, the pins constraint the members to translate in the 2D plane. Hence, the associate pin forces can be represented with two-components.
- When assembly is disassembled into their individual components, the pin joint forces form the action-reaction pairs in their respective FBDs.



Analysis of Frames and Machines...

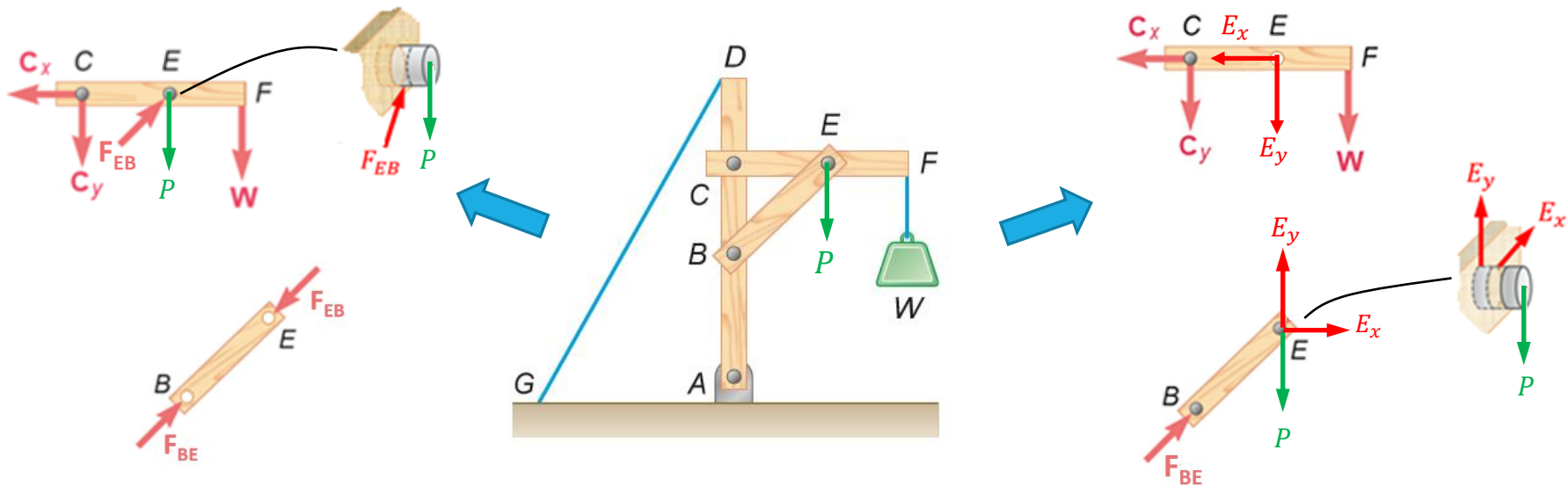
- Notice that the pin joints A , B , and C were attached to the vertical member, whereas for the pin joint E , it is removed altogether. Does it matter where the pins go to, or are they even needed to be included in the *FBDs*?



- In summary, when there is no other external force acting directly on the pin, other than the internal forces imposed on the connecting members, then it does not matter where the pin is placed.

Analysis of Frames and Machines...

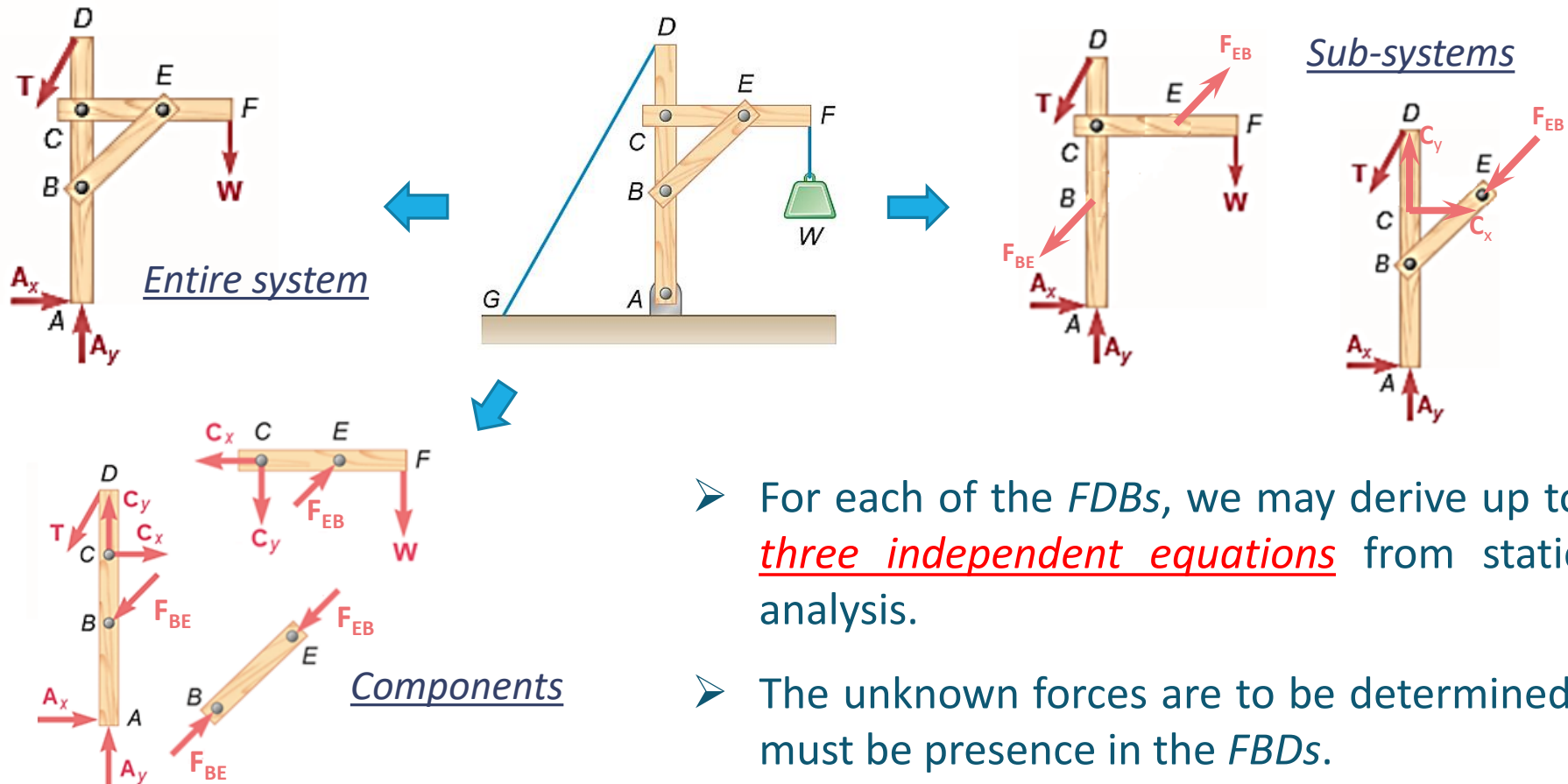
- Suppose there is an external force that is applied directly onto the pin of the system, such as load P on joint E .



- In this case, the pin must be included in your analysis, and it can only be associated with one of the components. In general, it should be placed in such a way that the internal force to be determined is revealed in the FBD and hence can be solved directly.

Analysis of Frames and Machines...

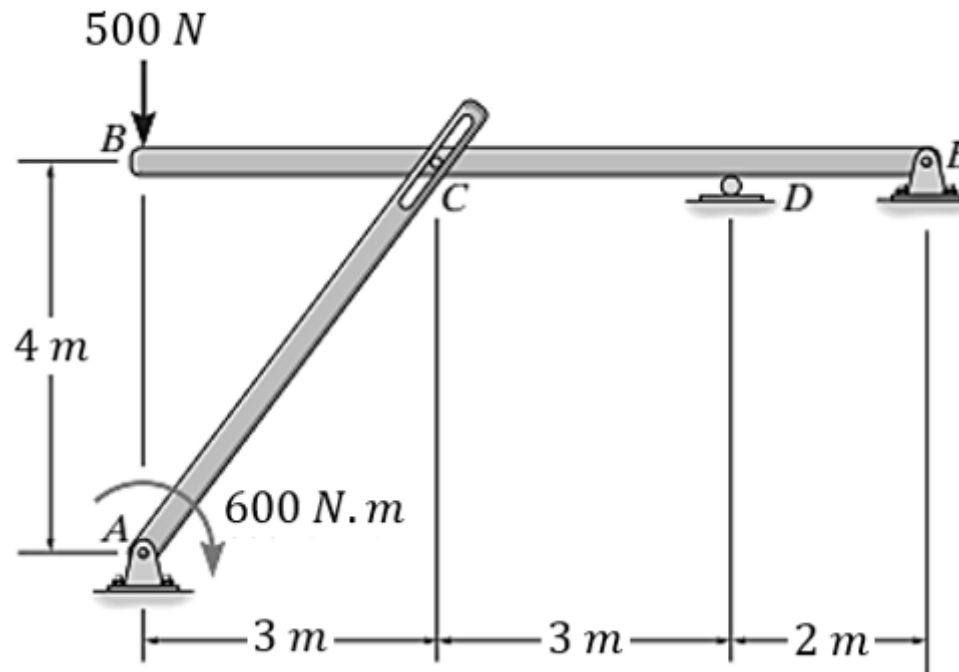
- The typical approach to solve the forces in the structure is to perform static analyses on: (i) the entire system, (ii) its components, and/or (iii) its sub-systems.



- For each of the *FDBs*, we may derive up to three independent equations from static analysis.
- The unknown forces are to be determined, must be presence in the *FDBs*.

Example 4.1

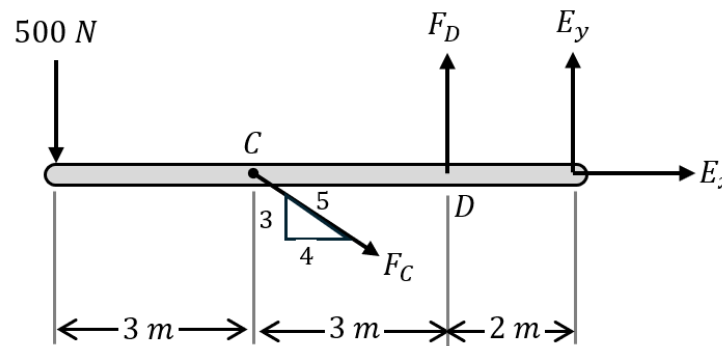
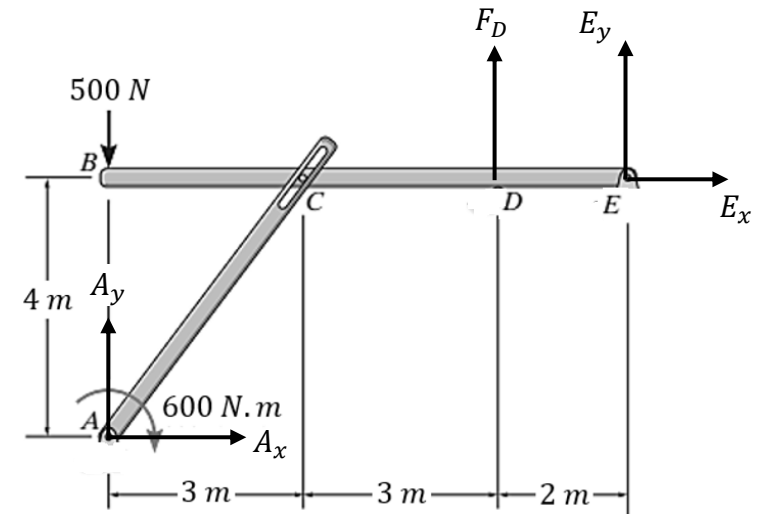
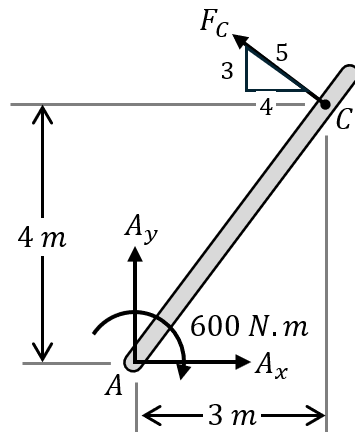
The two-member structure is connected at C by a pin, which is fixed to BDE and passes through the smooth slot in member AC . Determine the magnitude of the supporting forces at pin joints A and E .



Example 4.1...

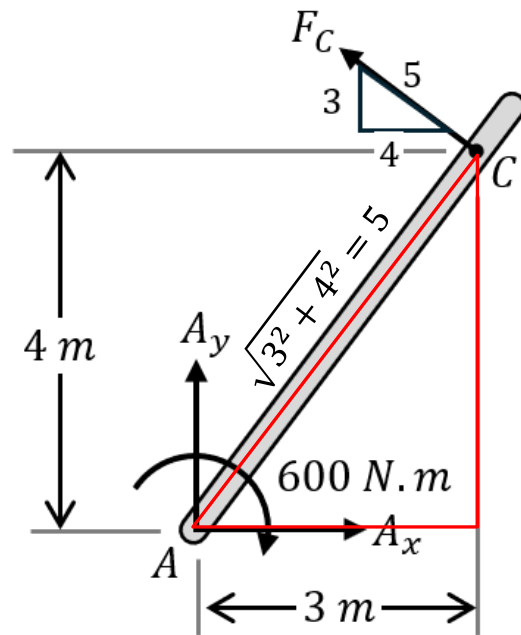
First, one should realize that because of the smooth slot, pin C exerts a normal force on the slot.

With that, three potential FBDs that can be defined for this example.



Example 4.1...

Among the three FBDs, using FBD of member AC can resolve all the unknown forces at A and C .



Taking moment about A , gives

$$-600 + F_C(5) = 0$$

$$\Rightarrow F_C = 120 \text{ N}$$

Since there are only two forces acting on member AC , the reaction force at A must be equal and opposite to F_C . Hence, magnitude of reaction force at A is **120 N**.

Example 4.1...

To determine the reaction at E , we can use either the *FBD* of member BCE or the entire assembly.

Consider member BCE , we have

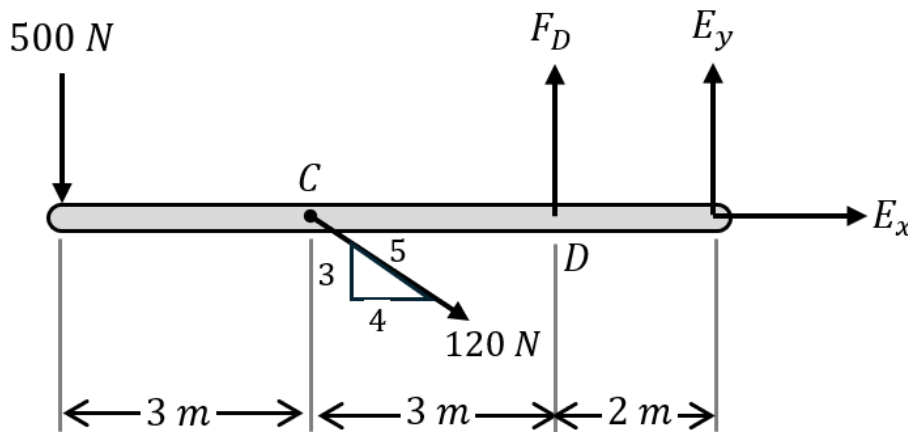
Taking moment about D , gives

$$500(6) + \left(\frac{3}{5}\right)(120)(3) + E_y(2) = 0$$

$$\Rightarrow E_y = -1608 \text{ N}$$

$$\sum F_x = 0, \left(\frac{4}{5}\right)(120) + E_x = 0$$

$$\Rightarrow E_x = -96 \text{ N}$$

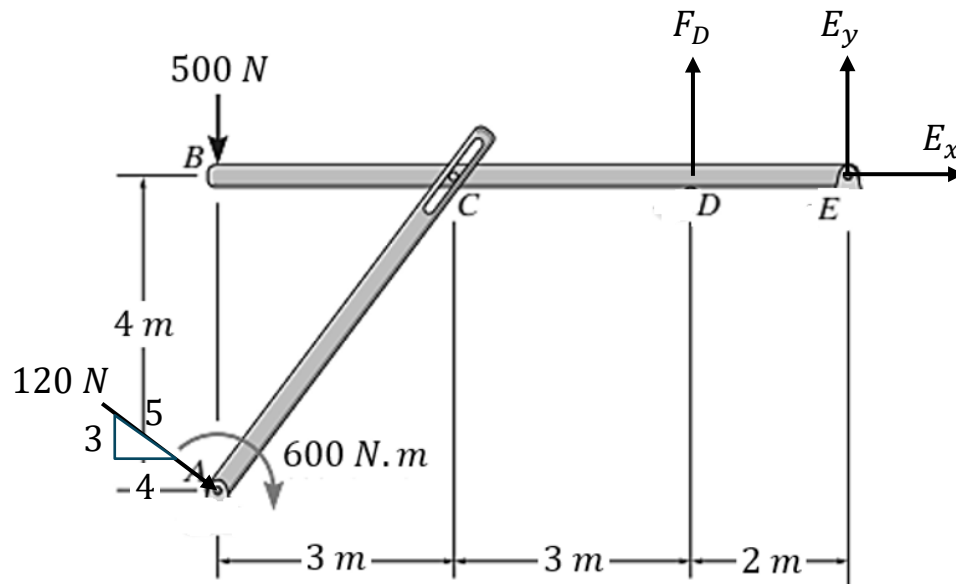


Hence, the magnitude of the reaction at E is

$$E = \sqrt{1608^2 + 96^2} = 1611 \text{ N}$$

Example 4.1...

Alternatively, by using the entire assembly



Taking moment about D , gives

$$-600 + 500(6) + \left(\frac{3}{5}\right)(120)(6) + \left(\frac{4}{5}\right)(120)(4) + E_y(2) = 0$$

$$\Rightarrow E_y = -1608 \text{ N}$$

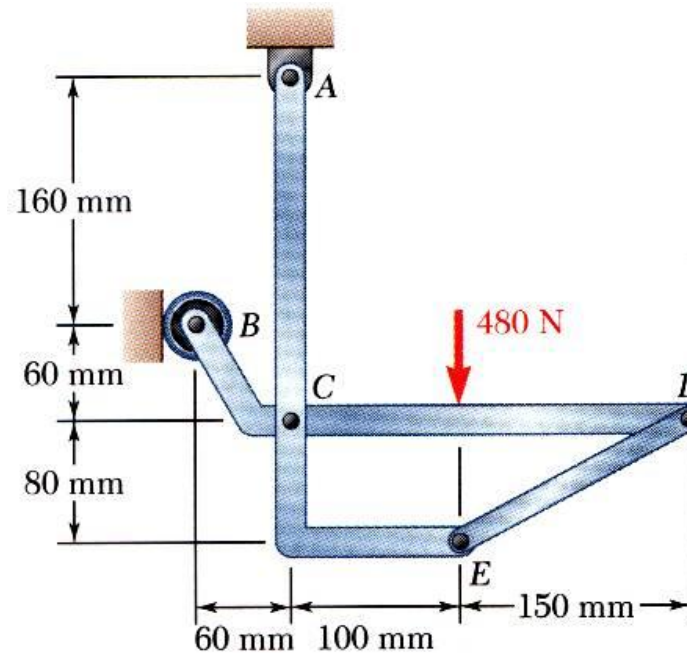
$$\sum F_x = 0, \left(\frac{4}{5}\right)(120) + E_x = 0$$

$$\Rightarrow E_x = -96 \text{ N}$$

Hence, the magnitude of the reaction at E is

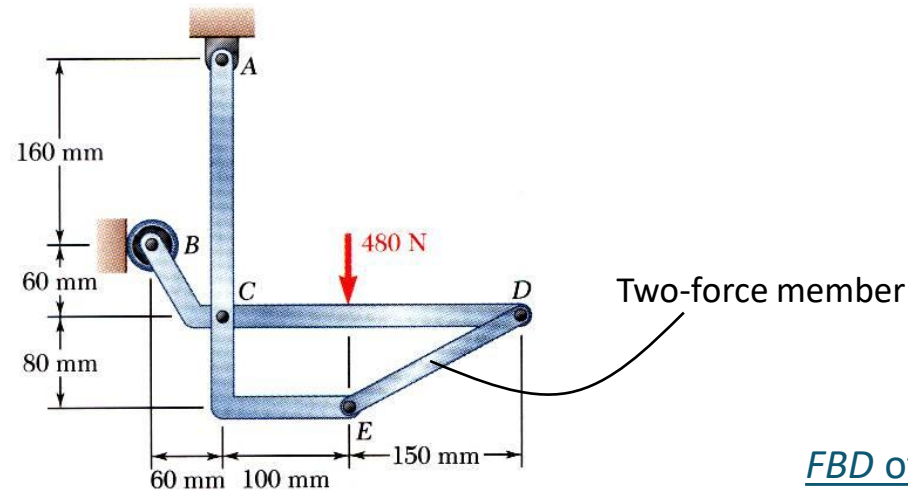
$$E = \sqrt{1608^2 + 96^2} = 1611 \text{ N}$$

Example 4.2

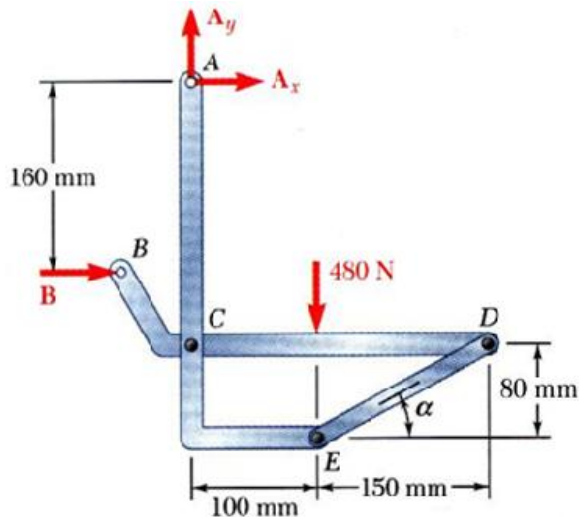


Members *ACE* and *BCD* are connected by a pin at *C* and by the link *DE*. For the loading shown, determine the force in link *DE* and the components of the force exerted at *C* on member *BCD*.

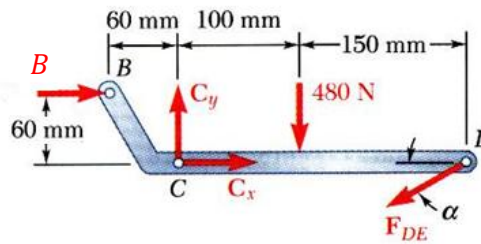
Example 4.2...



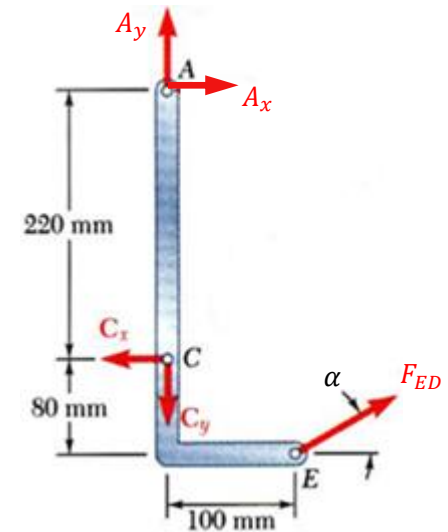
FBD of the entire frame



FBD of member BCD

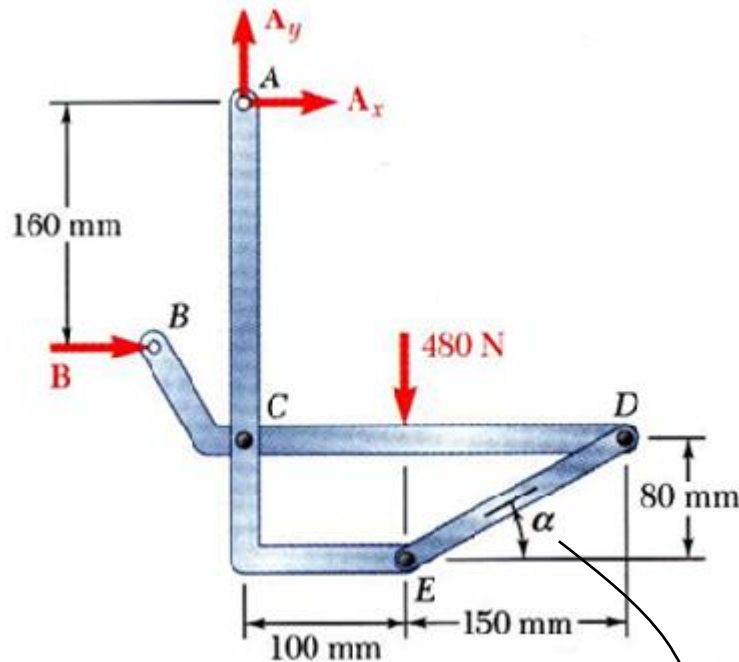


FBD of member ACE



Example 4.2...

FBD of the entire frame



$$\tan \alpha = \frac{80}{150} \Rightarrow \alpha = 28.07^\circ$$

Based on the *FBD* of the entire frame, we get

$$\sum M_A = 0, \quad -(480)(100) + B(160) = 0$$

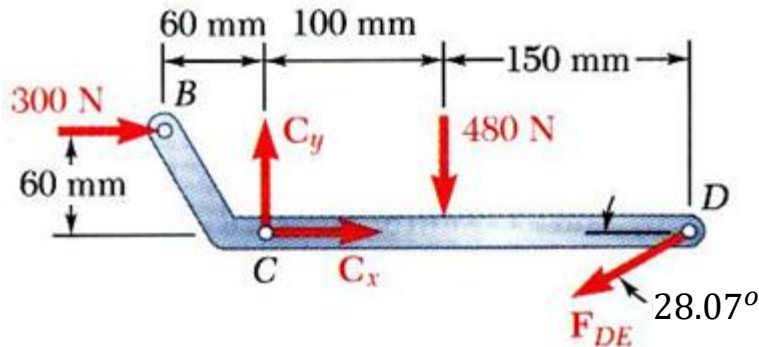
$$\Rightarrow B = 300 \text{ N}$$

$$\sum F_x = 0, \quad B + A_x = 0 \Rightarrow A_x = -300 \text{ N}$$

$$\sum F_y = 0, \quad A_y - 480 = 0 \Rightarrow A_y = 480 \text{ N}$$

Example 4.2...

FBD of member BCD



Consider member *BCD*. Since member *DE* is a two-force member, its force is along the member axis. Hence, there are only three unknown here, which can be solved using the static equilibrium.

$$\sum M_C = 0, \quad (F_{DE} \sin \alpha)(250) + (300)(60) + (480)(100) = 0$$

$$\Rightarrow F_{DE} = -561 \text{ N}$$

$$\sum F_x = 0, \quad C_x - F_{DE} \cos \alpha + 300 = 0 \Rightarrow C_x = -795 \text{ N}$$

$$\sum F_y = 0, \quad C_y - F_{DE} \sin \alpha - 480 = 0 \Rightarrow C_y = 216 \text{ N}$$

Example 4.2...

You can also perform analysis on member *ACE* instead, and/or use it to check your solution.

In this case, the force at joint *C* is drawn to be opposite to that define in the *FBD* of link *BCD*. This is because they correspond to the action and reaction forces.

$$\sum M_C = 0,$$

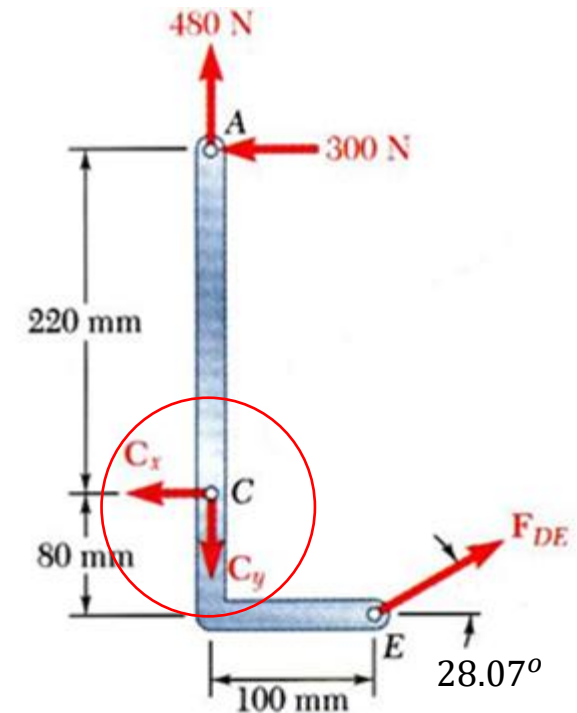
$$(F_{DE} \sin \alpha)(100) + (F_{DE} \cos \alpha)(80) + (300)(220) = 0$$

$$\Rightarrow F_{DE} = -561 \text{ N}$$

$$\sum F_x = 0, \quad -C_x + F_{DE} \cos \alpha - 300 = 0 \Rightarrow C_x = -795 \text{ N}$$

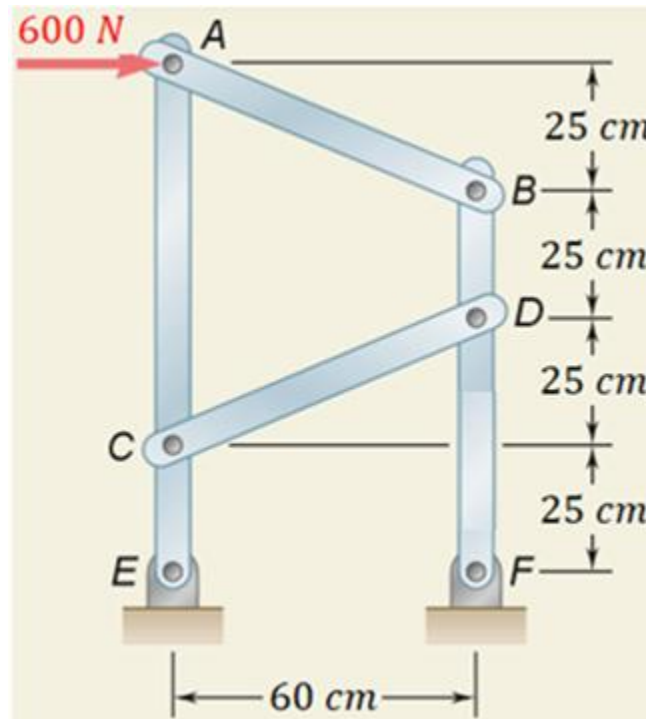
$$\sum F_y = 0, \quad -C_y + F_{DE} \sin \alpha + 480 = 0 \Rightarrow C_y = 216 \text{ N}$$

FBD of member *ACE*



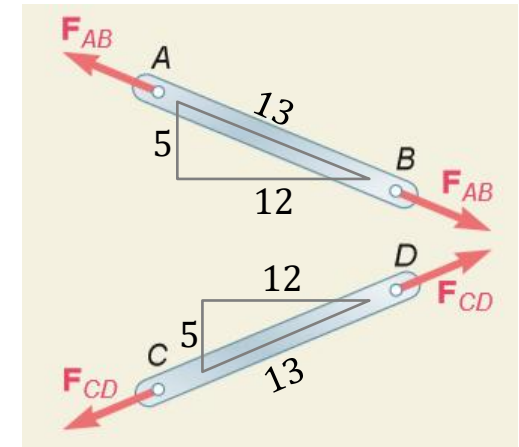
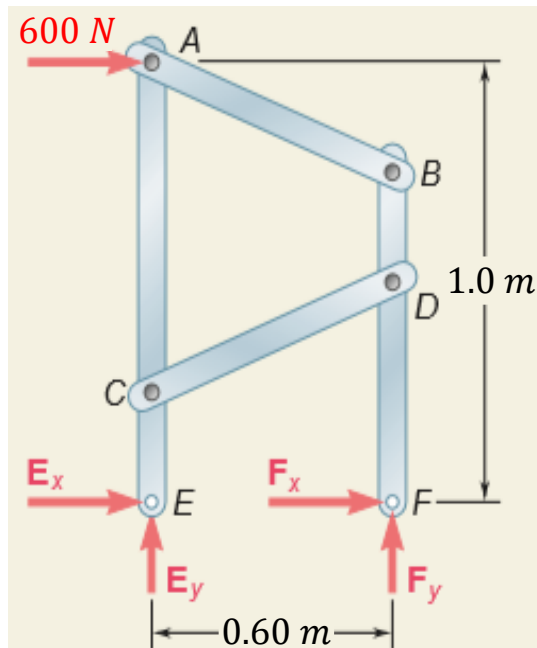
Example 4.3

A 600 N horizontal force is applied to pin A of the frame shown. Determine the forces in member AB and CD .



Example 4.3...

First, it should also be noted that members AB and CD are two-force members. Hence, the forces acting on them must be along the length of the members, which implies that the directions of the two forces are known.



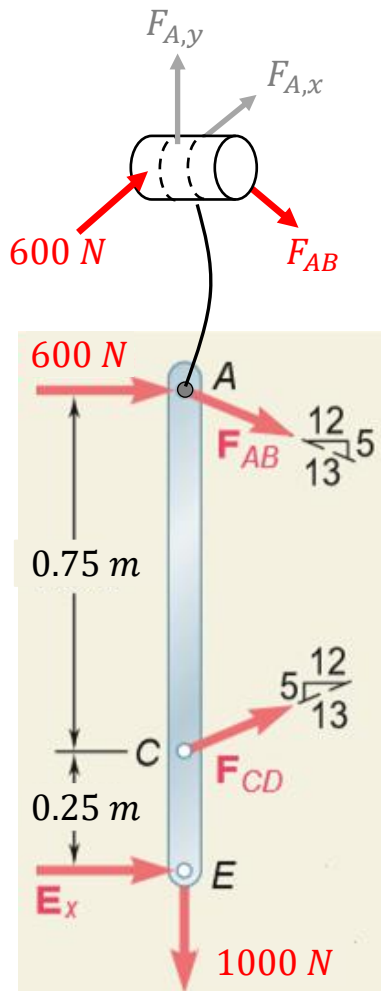
From the entire assembly, we have

$$\sum M_E = 0, \quad -(600)(1.0) + F_y(0.6) = 0 \Rightarrow F_y = 1000 \text{ N}$$

$$\sum F_y = 0, \quad E_y + F_y = 0 \Rightarrow E_y = -1000 \text{ N}$$

$$\sum F_x = 0, \quad E_x + F_x = -600$$

Example 4.3...



Now, consider the *FBD* of member ACE, with pin A attached to it. The static equilibrium gives the following equations.

Static equilibrium analysis gives

$$\sum M_E = 0, \quad (600)(1.0) + \left(\frac{12}{13}F_{AB}\right)(1.0) + \left(\frac{12}{13}F_{CD}\right)(0.25) = 0$$

$$\Rightarrow F_{AB} + 0.25F_{CD} = -650$$

$$\sum F_y = 0, \quad -\left(\frac{5}{13}F_{AB}\right) + \left(\frac{5}{13}F_{CD}\right) - 1000 = 0$$

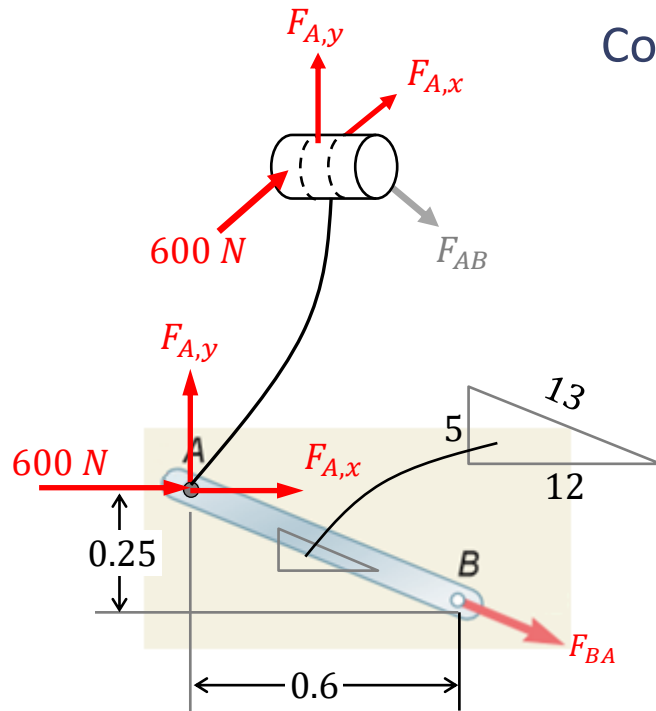
$$\Rightarrow F_{CD} - F_{AB} = 2600$$

Solving the two equations simultaneously, we get

$$F_{AB} = -1040 \text{ N}, \quad F_{CD} = 1560 \text{ N}$$

Example 4.3...

Alternatively, suppose pin A is attached to member AB . In this case, the 600 N force will be imposed on the *FBD* of AB .



Consider force equilibrium of member AB

$$F_{A,x} + \frac{12}{13}F_{BA} + 600 = 0 \Rightarrow F_{A,x} = -600 - \frac{12}{13}F_{BA}$$

$$F_{A,y} - \frac{5}{13}F_{BA} = 0 \Rightarrow F_{A,y} = \frac{5}{13}F_{BA}$$

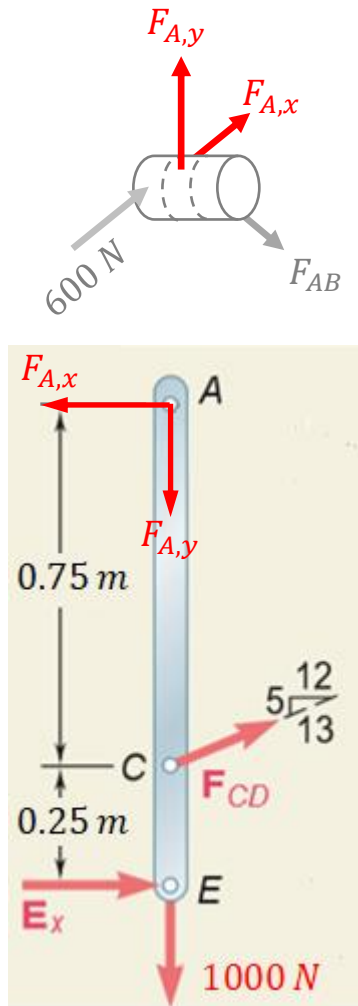
$$\sum M_B = 0, \quad -(600 + F_{A,x})(0.25) - F_{A,y}(0.6) = 0$$

$$\Rightarrow F_{A,x} + 2.4F_{A,y} = -600$$

This can be derived from the two force equations, and hence it is not an independent equation.

Member AB is a two-force member. This means that the resultant force at pin A must be equal and opposite to F_{BA} .

Example 4.3...



$$\sum M_E = 0, \quad F_{A,x}(1) - \frac{12}{13}F_{CD}(0.25) = 0 \Rightarrow F_{A,x} = \frac{3}{13}F_{CD}$$

$$\sum F_x = 0, \quad -F_{A,x} + \frac{12}{13}F_{CD} + E_x = 0$$

$$\sum F_y = 0, \quad -F_{A,y} + \frac{5}{13}F_{CD} - 1000 = 0$$

Substitute $F_{A,x} = \frac{3}{13}F_{CD}$ into $F_{A,x} = -600 - \frac{12}{13}F_{AB}$, gives

$$\frac{3}{13}F_{CD} = -600 - \frac{12}{13}F_{AB} \Rightarrow 3F_{CD} + 12F_{AB} = -7800$$

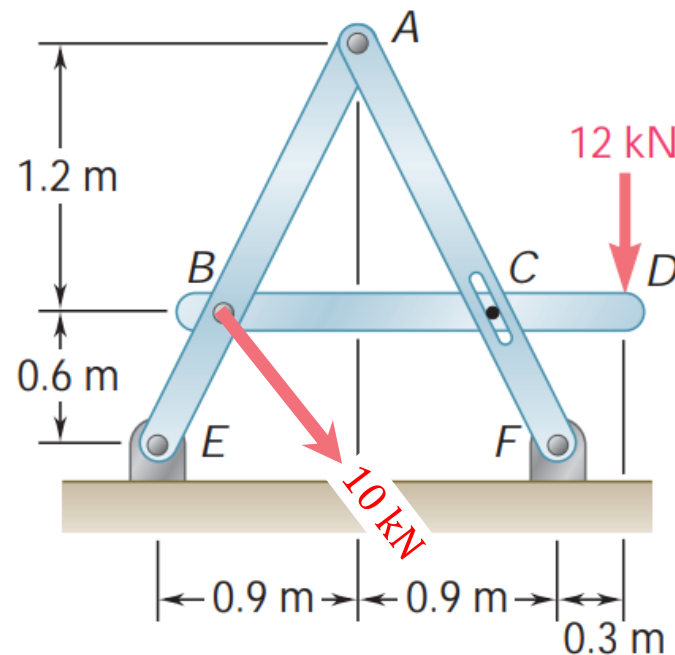
Substitute $F_{A,y} = \frac{5}{13}F_{AB}$ into $-F_{A,y} + \frac{5}{13}F_{CD} - 1000 = 0$, gives

$$-\frac{5}{13}F_{AB} + \frac{5}{13}F_{CD} - 1000 = 0 \Rightarrow -F_{AB} + F_{CD} = 2600$$

Solving the equations, gives $F_{AB} = -1040 \text{ N}$ and $F_{CD} = 1560 \text{ N}$.

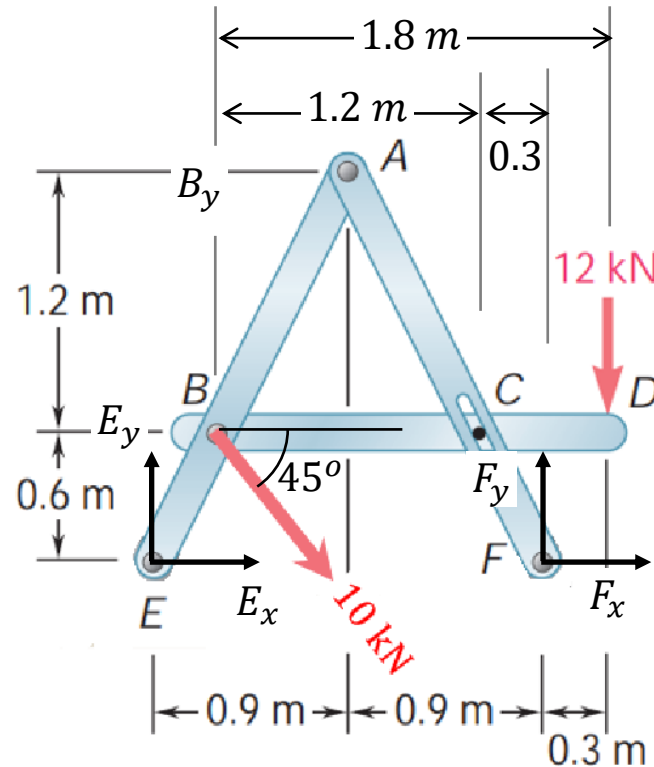
Example 4.4

For the frame and loading shown, determine the all the pin forces acting on member *ABE*. Note that the pin at *C* can slide freely along the slot.



Example 4.4...

Consider the entire frame.



Taking moment about F , gives

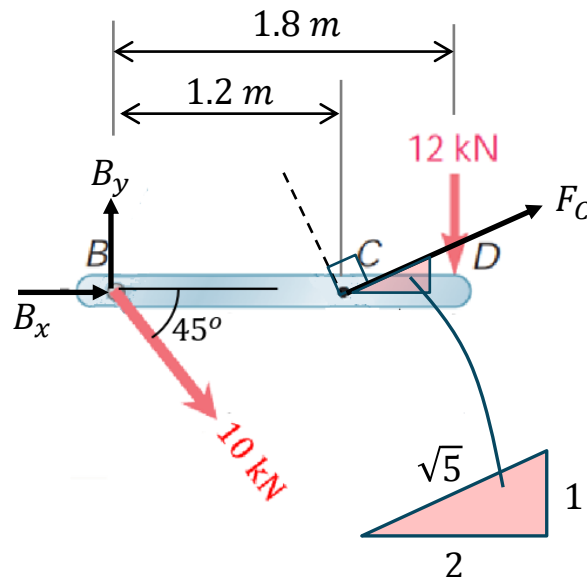
$$-E_y(1.8) + (10\sin 45^\circ)(1.5) - (10\cos 45^\circ)(0.6) - 12(0.3) = 0$$

$$\Rightarrow E_y = 1.536 \text{ kN}$$

Example 4.4...

Next, consider member BCD , with pin B in place. This implies that reaction force at B corresponds to the pin force imposed by member ABE .

Note also that the pin force at C can slide freely along the slot, which means that there is only normal force and is perpendicular to AC .



Taking moment about B , we have

$$\left(\frac{1}{\sqrt{5}} F_C \right) 1.2 - 12(1.8) = 0 \Rightarrow F_C = 40.25 \text{ kN}$$

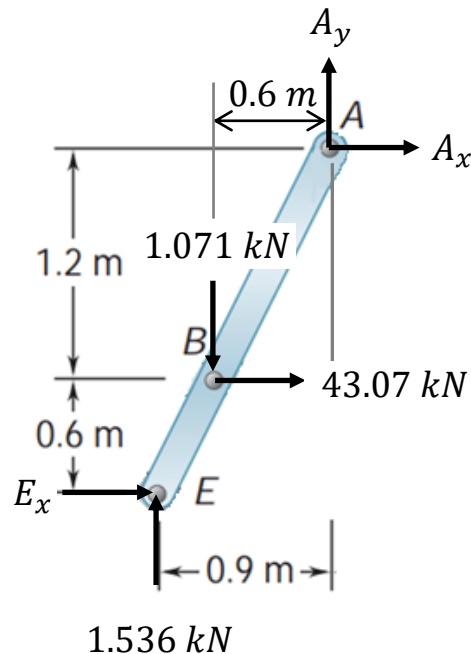
And force equations, give

$$B_x + 10\cos 45 + \frac{2}{\sqrt{5}} F_C = 0 \Rightarrow B_x = -43.07 \text{ kN}$$

$$B_y - 10\sin 45 + \frac{1}{\sqrt{5}} F_C - 12 = 0 \Rightarrow B_y = 1.071 \text{ kN}$$

Example 4.4...

Finally, performing static analysis of member *ABE*, gives



$$\sum M_A = 0, \quad E_x(1.8) - 1.536(0.9) + 1.071(0.6) + 43.07(1.2) = 0$$

$$\Rightarrow E_x = -\mathbf{28.30 \text{ kN}}$$

$$\sum F_x = 0, \quad A_x + 43.07 - 28.30 = 0$$

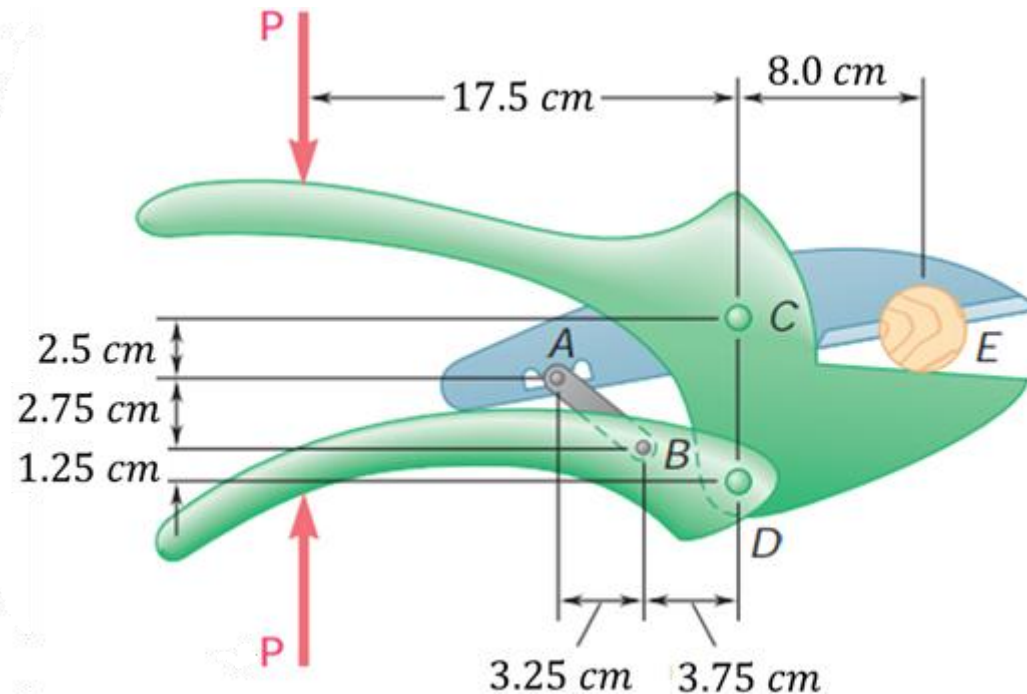
$$\Rightarrow A_x = -\mathbf{14.77 \text{ kN}}$$

$$\sum F_y = 0, \quad A_y - 1.071 + 1.536 = 0$$

$$\Rightarrow A_y = -\mathbf{0.465 \text{ kN}}$$

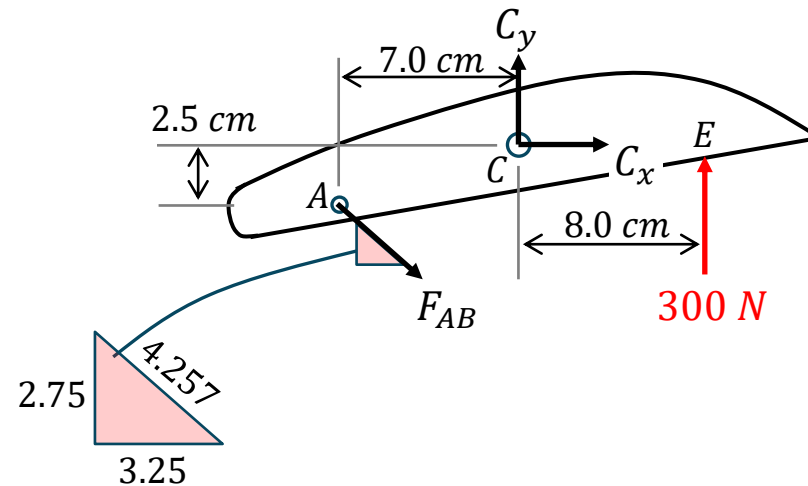
Example 4.5

The compound-lever pruning shears shown can be adjusted by placing pin A at various ratchet positions on blade ACE . Knowing that 300 N vertical forces at E are required to complete the pruning of a small branch, determine the magnitude of force P that must be applied to the handles to complete the cutting task.



Example 4.5...

Consider the cutting blade *ACE*.



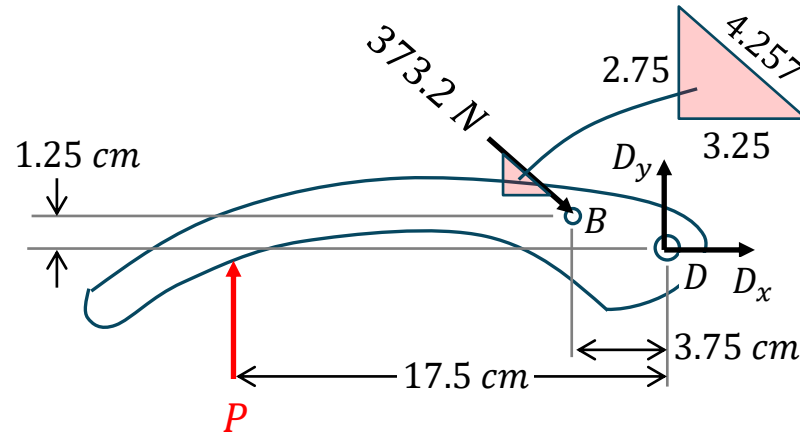
Taking moment about C, we have

$$300(8) + \left(\frac{2.75}{4.257} F_{AB} \right) (7) + \left(\frac{3.25}{4.257} F_{AB} \right) (2.5) = 0$$

$$\Rightarrow F_{AB} = -373.2 \text{ N}$$

Example 4.5...

Finally, consider the lower handle *ACE*.



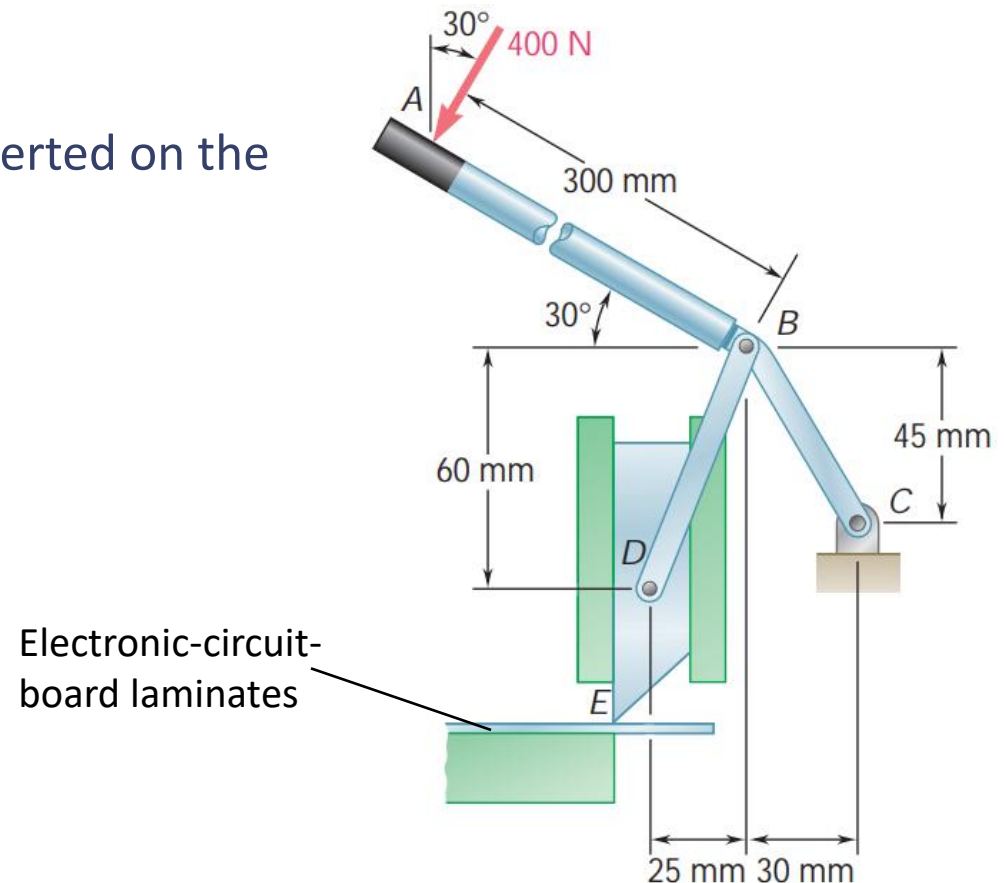
The moment equation about *D*, gives

$$\begin{aligned}
 -P(17.5) + \left(\frac{2.75}{4.257} (373.2) \right) (3.75) - \left(\frac{3.25}{4.257} (373.2) \right) (1.25) &= 0 \\
 \Rightarrow P &= \mathbf{31.31 \text{ N}}
 \end{aligned}$$

Example 4.6

The shear mechanism shown is used to cut and trim electronic-circuit-board laminates. Assuming the cutting blade can slide frictionless along the guiding plates, determine:

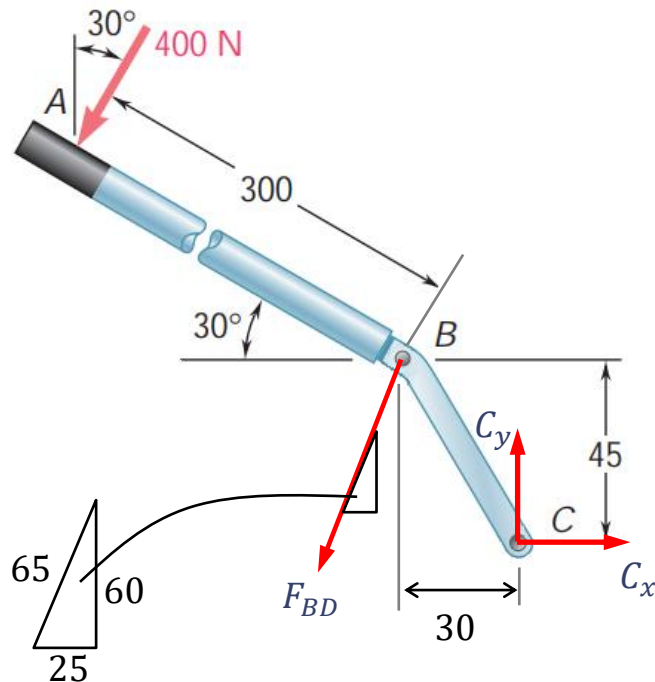
- the reaction at C , and
- the corresponding vertical force exerted on the shearing blade at E .



Example 4.6...

(a) First, it should be noted that BD is a two-force member.

Consider member ABC



Based on the geometry, the moment arm vectors for the forces are:

$$\vec{r}_{B/C} = -0.03\vec{i} + 0.045\vec{j}$$

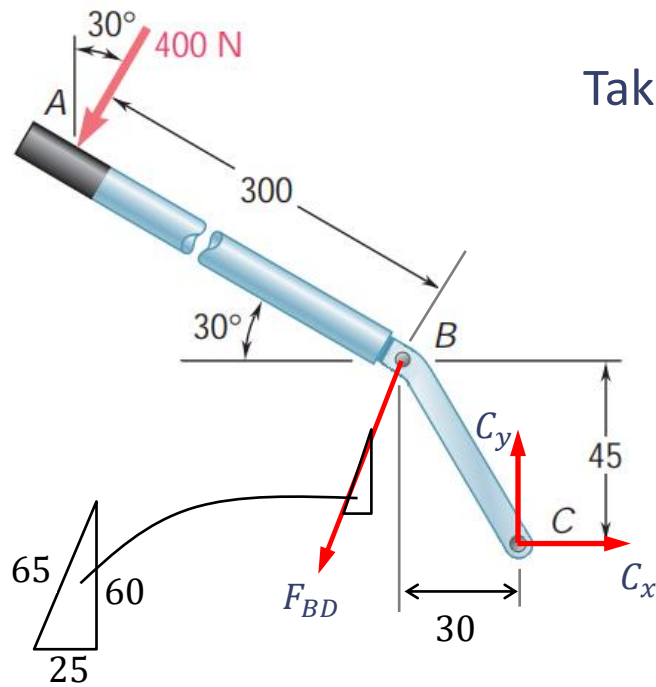
$$\begin{aligned}\vec{r}_{A/C} &= (-0.03 - 0.3\cos 30^\circ)\vec{i} + (0.045 + 0.3\sin 30^\circ)\vec{j} \\ &= -0.2898\vec{i} + 0.195\vec{j}\end{aligned}$$

And the corresponding forces are:

$$\vec{F}_{BD} = F_{BD} \left(-\frac{25}{65}\vec{i} - \frac{60}{65}\vec{j} \right)$$

$$\begin{aligned}\vec{F}_{400} &= 400(-\sin 30^\circ\vec{i} - \cos 30^\circ\vec{j}) \\ &= -200\vec{i} - 346.4\vec{j}\end{aligned}$$

Example 4.6...



Taking moment about C, gives

$$\begin{aligned}\sum M_C &= 0, \quad \vec{r}_{B/C} \times \vec{F}_{BD} + \vec{r}_{A/C} \times \vec{F}_{400} = 0 \\ &\Rightarrow \left(-0.03 \left(-\frac{60}{65} F_{BD} \right) - 0.045 \left(-\frac{25}{65} F_{BD} \right) \right) \\ &\quad + (-0.2898(-346.4) - 0.195(-200)) = 0 \\ &\Rightarrow F_{BD} = -3097.6 \text{ N}\end{aligned}$$

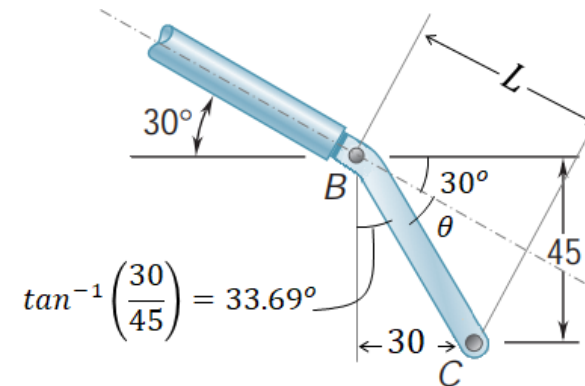
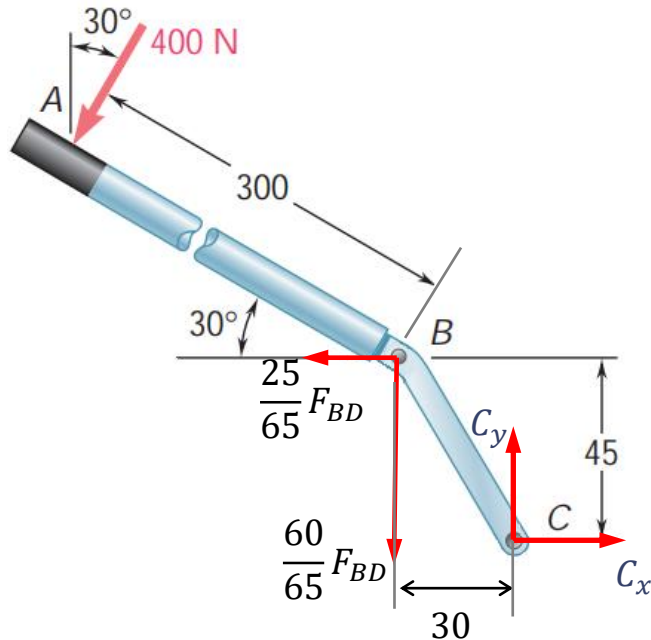
Next, performing static analysis on the bar, gives

$$\sum F_x = 0, \quad C_x - \frac{25}{65}(-3097.6) - 200 = 0 \Rightarrow C_x = -\mathbf{991.4 \text{ N}}$$

$$\sum F_y = 0, \quad C_y - 346.4 - \left(\frac{60}{65} \right)(-3097.6) = 0 \Rightarrow C_y = -\mathbf{2512.9 \text{ N}}$$

Example 4.6...

Alternatively, you can also determine the moment equation as follows:



$$\theta = 90^\circ - 30^\circ - 33.69^\circ = 26.31^\circ$$

$$BC = \sqrt{45^2 + 30^2} = 54.08 \text{ mm}$$

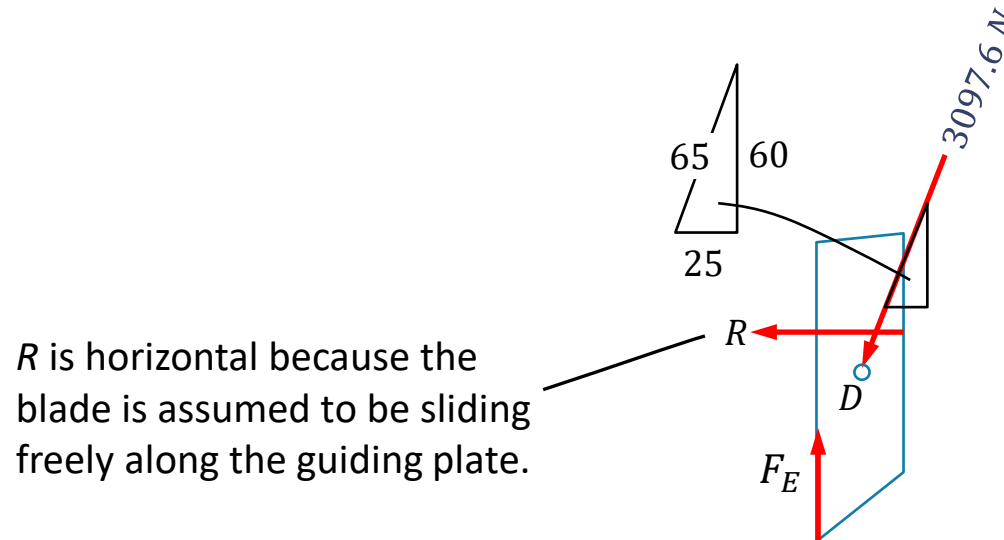
$$L = BC \cos \theta = 48.48 \text{ mm}$$

$$\sum M_C = 0, \quad \frac{25}{65} F_{BD} (0.045) + \frac{60}{65} F_{BD} (0.03) + 400 (0.3 + 0.04848) = 0$$

$$\Rightarrow F_{BD} = -3097.6 \text{ N}$$

Example 4.6...

Finally, consider the cutting blade.

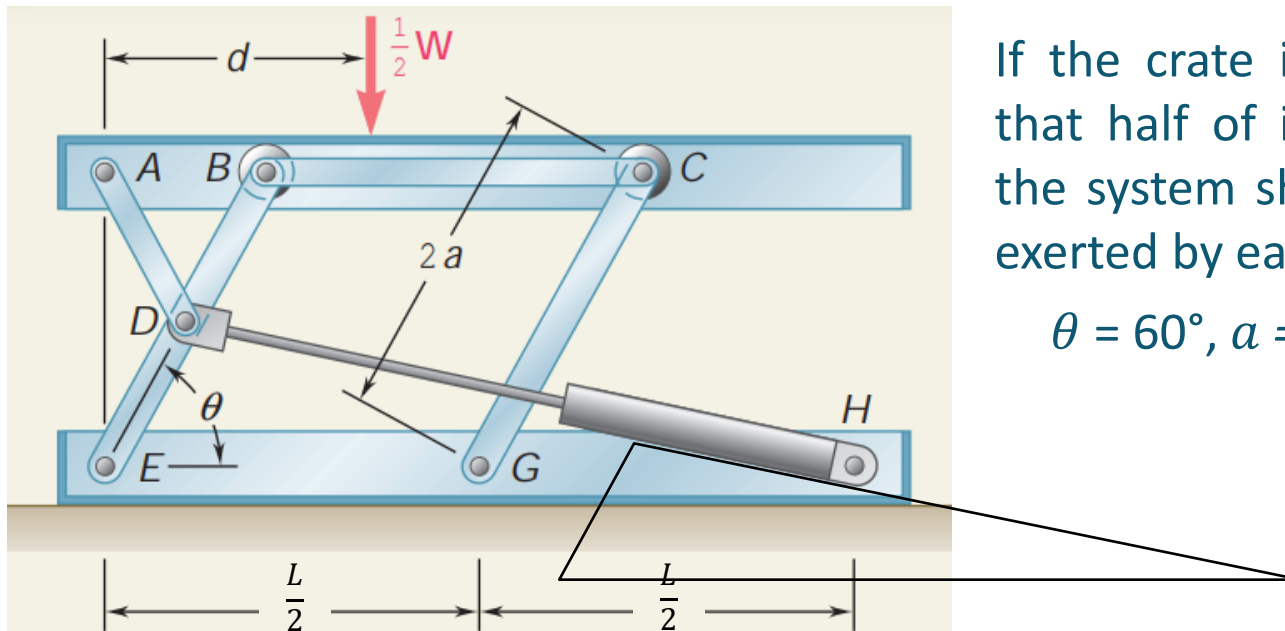


And by the force in the y -direction gives

$$F_E - \frac{60}{65} (3097.6) = 0 \Rightarrow F_E = \mathbf{2859\text{ N}}$$

Example 4.7

A hydraulic-lift table is used to raise a 1000 *kg* crate. It consists of a platform and two identical linkages on which hydraulic cylinders exert equal forces. Members *EDB* and *CG* are each of length $2a$, and member *AD* is pinned to the midpoint of *EDB*.



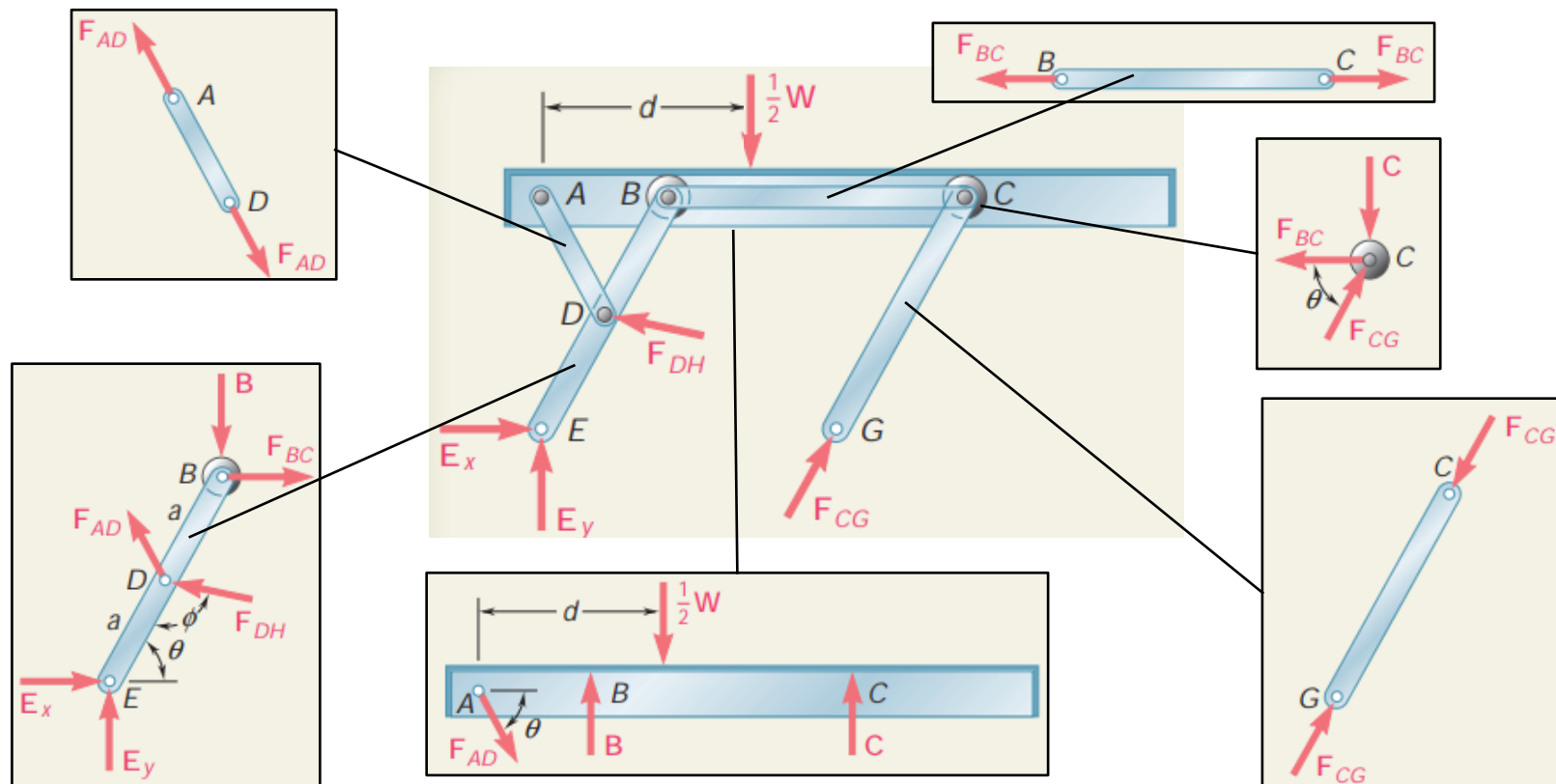
If the crate is placed on the table, so that half of its weight is supported by the system shown, determine the force exerted by each cylinder for:

$$\theta = 60^\circ, a = 0.70 \text{ m}, \text{ and } L = 3.20 \text{ m}.$$

Example 4.7...

Consider the mechanism of the hydraulic-lift table as shown. It involves four unknowns in the *FBD*, which makes it not suitable to determine the cylinder force.

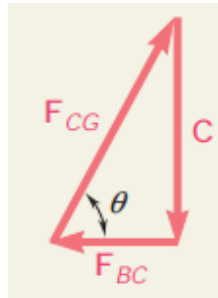
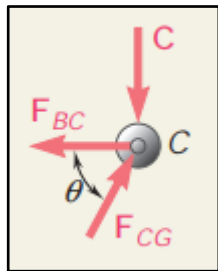
In this case, we must dismember the mechanism into its components, as shown.



Example 4.7...

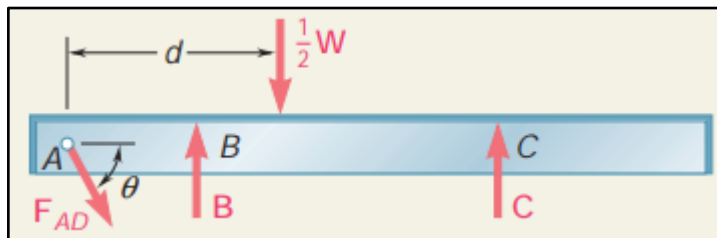
Among the *FBDs* of the components, the two-force members *AD*, *BC* and *CG* *only* define the directions of the forces.

Only member *BDE* contains the cylinder force and hence must be used to solve the problem. However, it contains other unknowns that required the analyses of platform *ABC* and roller *C*.



Consider roller *C*,

$$\frac{C}{F_{CB}} = \tan\theta \Rightarrow F_{CB} = 0.57735C$$



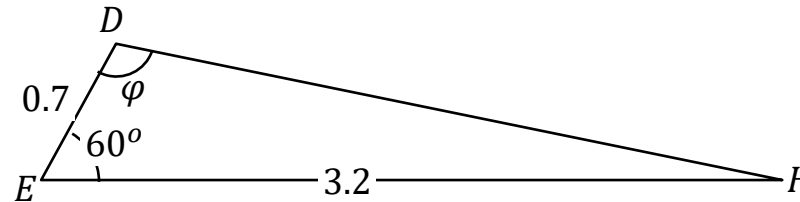
Consider platform *ABC*,

$$F_{AD} = 0, \quad \text{and} \quad B + C = \frac{W}{2} = 500g$$

Example 4.7...

Now, consider member *BDE*.

Based on the geometry, i.e. triangle *DEH*,



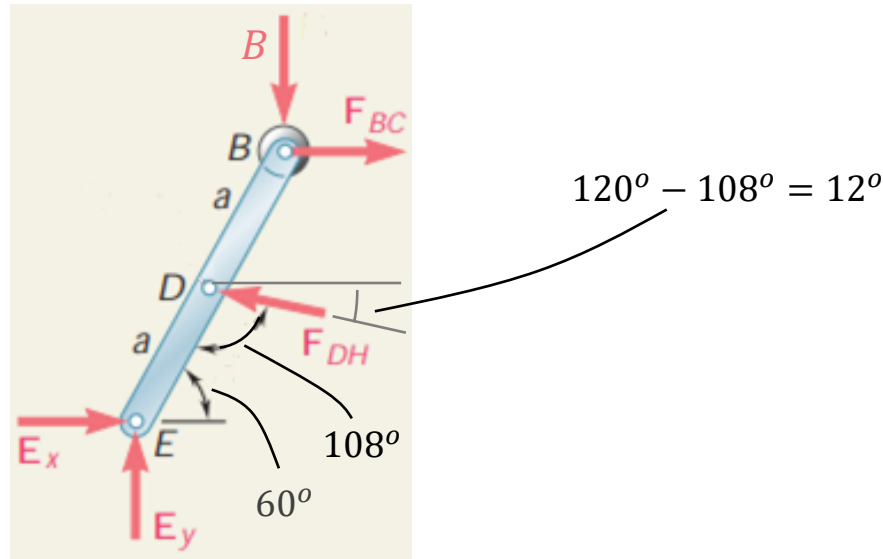
Using cosine rule, we have

$$DH^2 = 0.7^2 + 3.2^2 - 2(0.7)(3.2)\cos 60^\circ \Rightarrow DH = 2.914$$

And then sine rule,

$$\frac{\sin \varphi}{3.2} = \frac{\sin 60^\circ}{2.914} \Rightarrow \varphi = 72.0^\circ \text{ or } 108^\circ$$

Example 4.7...



Finally, taking moment about E , gives

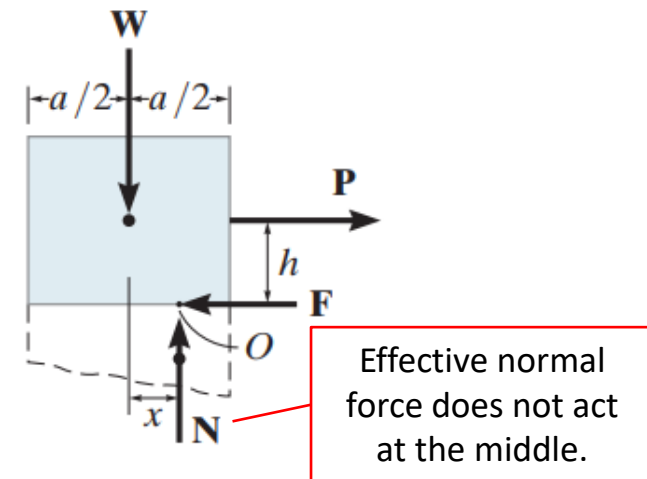
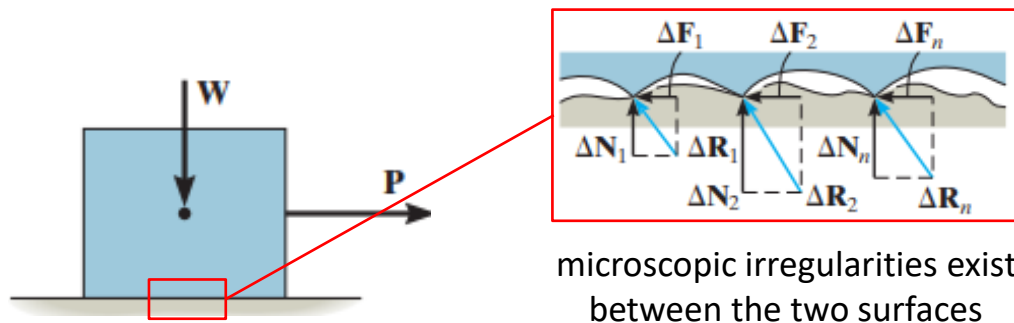
$$-B(2a\cos 60^\circ) - F_{BC}(2a\sin 60^\circ) + F_{DH}\cos 12^\circ(as\sin 60^\circ) + F_{DH}\sin 12^\circ(acos 60^\circ) = 0$$

$$\Rightarrow 0.9511F_{DH} = B + C = 500g$$

$$\Rightarrow F_{DH} = \mathbf{5157\text{ N}}$$

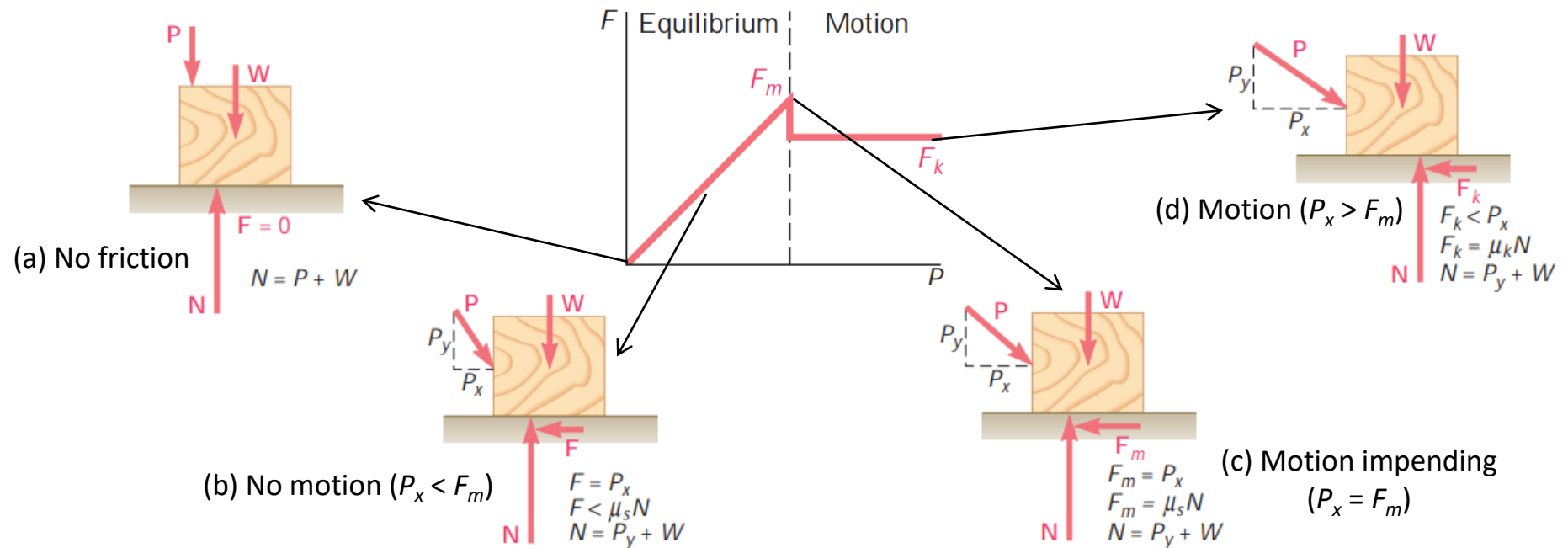
Friction

- So far, we have assumed that all contact surfaces are frictionless, which:
 - give rise to only normal force at the contact surfaces, and
 - the two surfaces can slide freely over each other.
- In reality, dry friction (Coulomb) will always develop at the contact interfaces when they attempt to move relative to one another.
- It is tangential to the contact surface that resist motion, and the magnitude is limited, depending on the surface conditions at the interface.



Friction...

- Experimental evidence shows the frictional force is directly proportional to the normal component of the reaction at the surface.



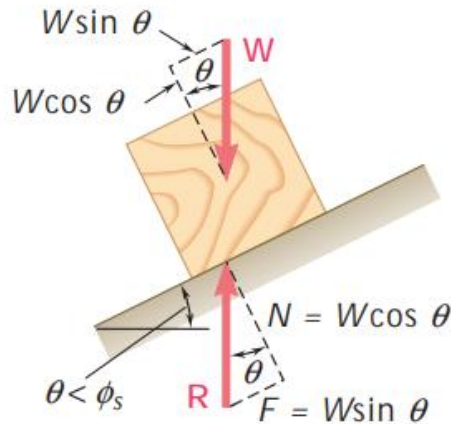
- Based on this plot, we can define

$$F_m = \mu_s N, \quad \text{and} \quad F_k = \mu_k N.$$

where μ_s and μ_k denote the static and kinetic friction coefficients, respectively.

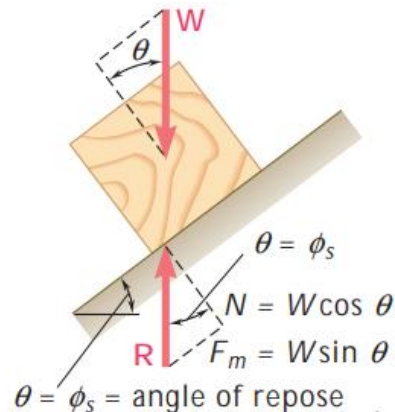
Friction...

- The static friction coefficient μ_s can also be estimated from the inclination test as follows:



The ratio of the friction to normal force is,

$$\frac{F}{N} = \frac{W \sin \theta}{W \cos \theta} = \tan \theta$$

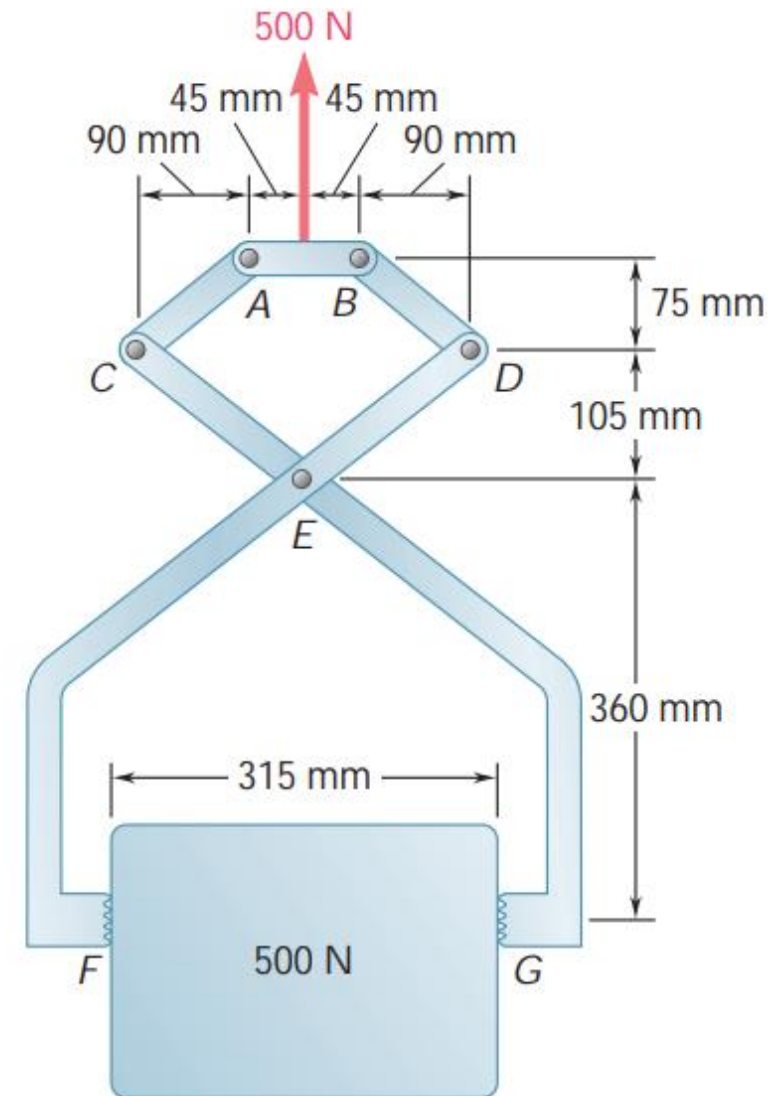


If the inclination angle is further increased until it just starts to slide downward. At this inclination angle, ϕ_s , the friction force just pass its maximum at F_m . Hence, the static friction coefficient is

$$\frac{F_m}{N} = \mu_s = \tan \phi_s$$

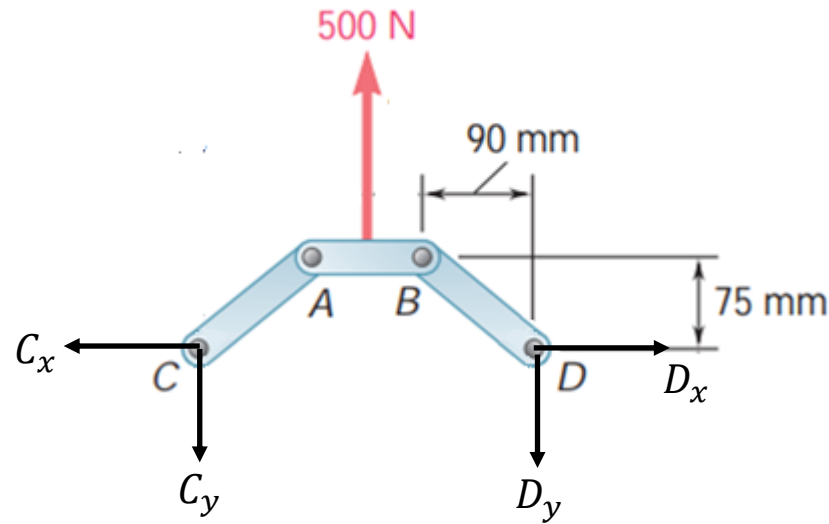
Example 4.8

A 500- N concrete block is to be lifted by the pair of tongs as shown in the figure. Determine the smallest allowable value of the coefficient of static friction at the contact surfaces between the block and the tongs at F and G .



Example 4.8...

Consider the sub-assembly *CABD*.



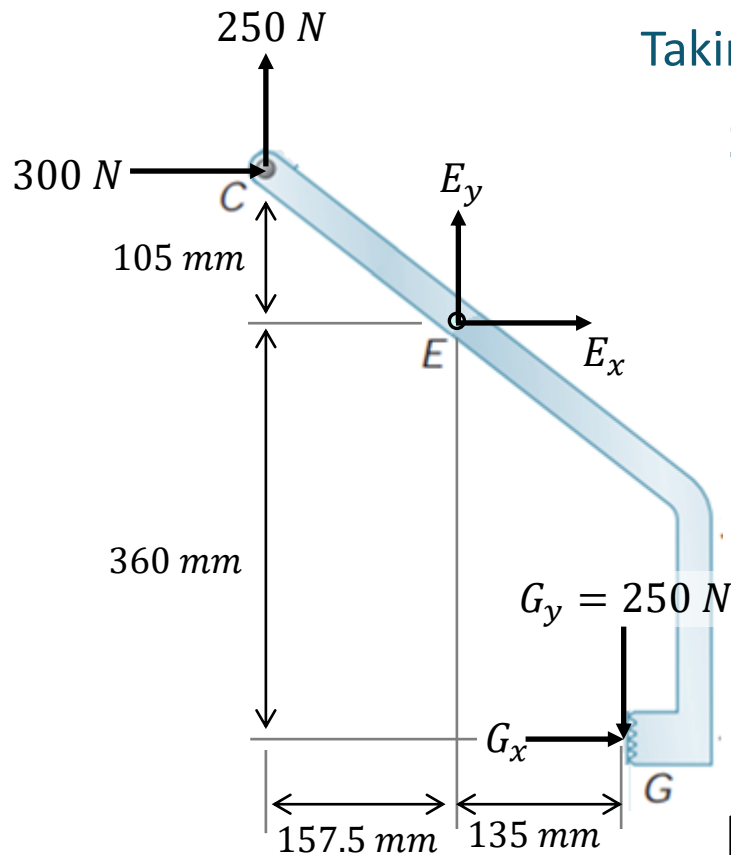
By symmetry, we know that $C_x = D_x$, and $C_y = D_y$. This means that the vertical components will each bear half of the applied load, i.e. $C_y = D_y = 250\text{ N}$.

And notice that links *AC* and *BD* are two-force members. Hence, this means

$$\frac{C_x}{C_y} = \frac{90}{75} \Rightarrow C_x = 300\text{ N}$$

Example 4.8...

Next, consider the tong link *CEG*.



Taking moment about *E*, we have

$$300(105) + 250(135) + 250(157.5) - G_x(360) = 0$$

$$\Rightarrow G_x = 290.625 \text{ N}$$

Finally, equating the friction force G_y to the maximum friction force, i.e. $G_y = \mu_s G_x$, we get the minimum static frictional coefficient to be

$$\mu_s = \frac{G_y}{G_x} = \frac{250}{290.625} = \mathbf{0.860}$$

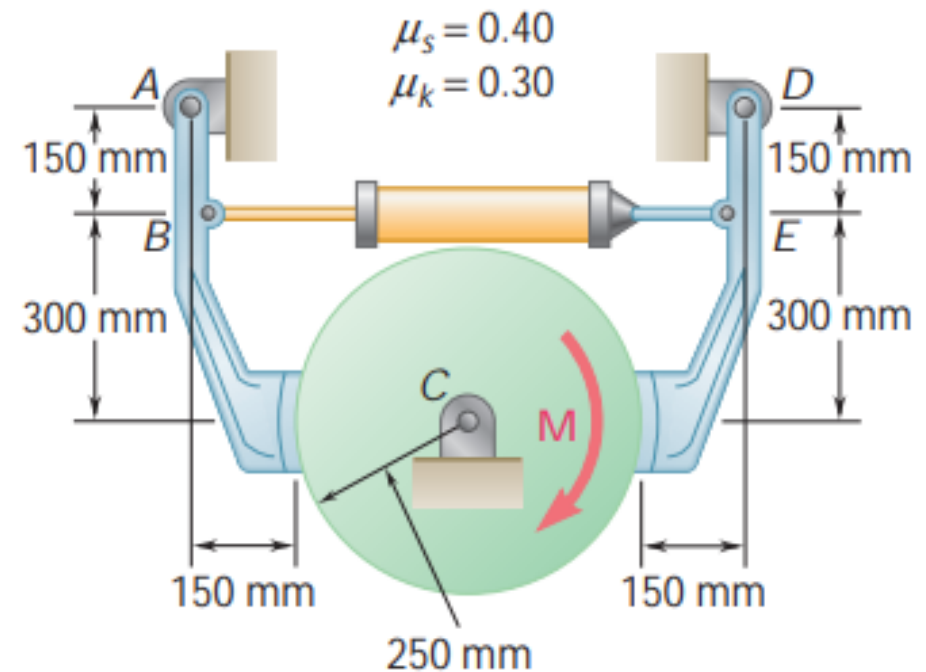
This comes from force equilibrium of the block.

Example 4.9

The hydraulic cylinder exerts braking forces on the drum as shown.

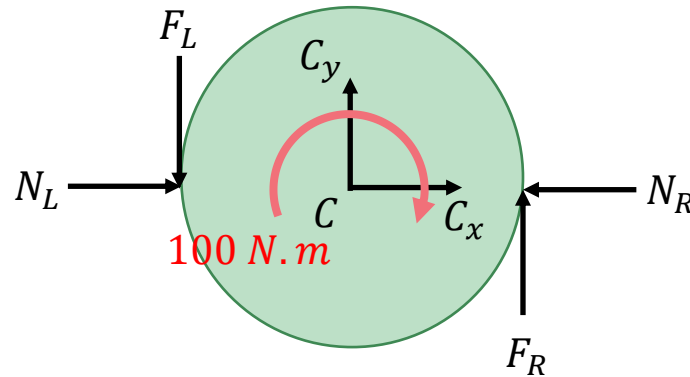
(a) Given that a couple M of magnitude 100 N.m is applied to the drum. Determine the smallest force that must be exerted by the hydraulic cylinder if the drum is not to rotate.

(b) Suppose the cylinder force is 3 kN . Determine the magnitude of the couple M required to rotate the drum clockwise at a constant speed.



Example 4.9...

(a) Consider the *FBD* of the drum.



Taking moment about C, we have

$$(0.25)(F_L + F_R) - 100 = 0$$

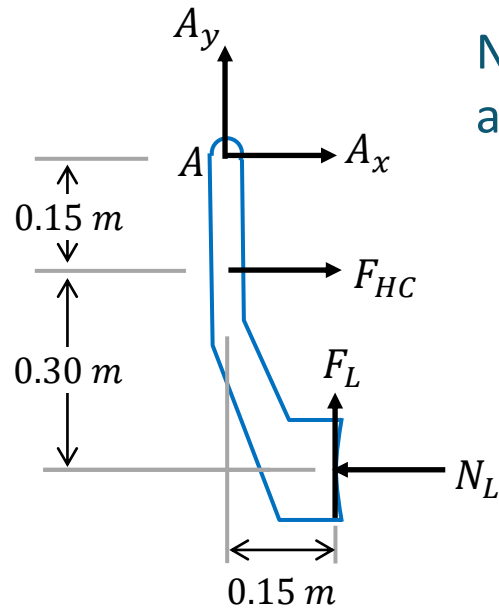
$$\Rightarrow F_L + F_R = 400 \text{ N}$$

In this case, we can assume that the drum is at impending motion, which implies

$$F_L = \mu_s N_L = 0.4 N_L \Rightarrow N_L = 2.5 F_L$$

$$F_R = 0.4 N_R \Rightarrow N_R = 2.5 F_R$$

Example 4.9



Next, consider the *FBD* of the left arm. And taking moment about A, gives

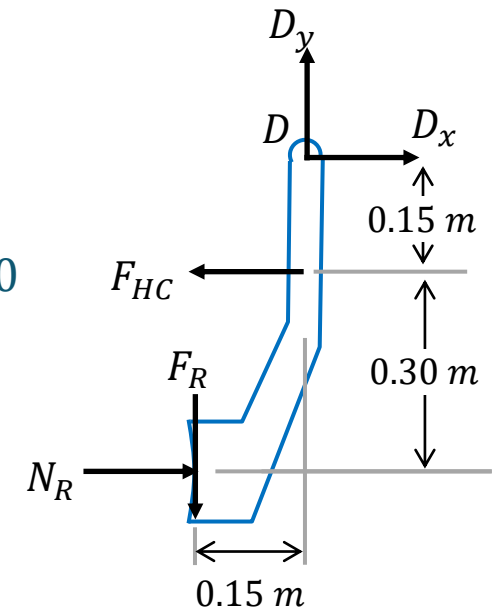
$$\begin{aligned}
 F_{HC}(0.15) + F_L(0.15) - N_L(0.45) &= 0 \\
 \Rightarrow F_{HC}(0.15) + F_L(0.15) - (2.5F_L)(0.45) &= 0 \\
 \Rightarrow F_L &= 0.1538F_{HC}
 \end{aligned}$$

Similarly for the right arm, we have

$$\begin{aligned}
 F_{HC}(0.15) - F_R(0.15) - N_R(0.45) &= 0 \\
 \Rightarrow F_R &= 0.1177F_{HC}
 \end{aligned}$$

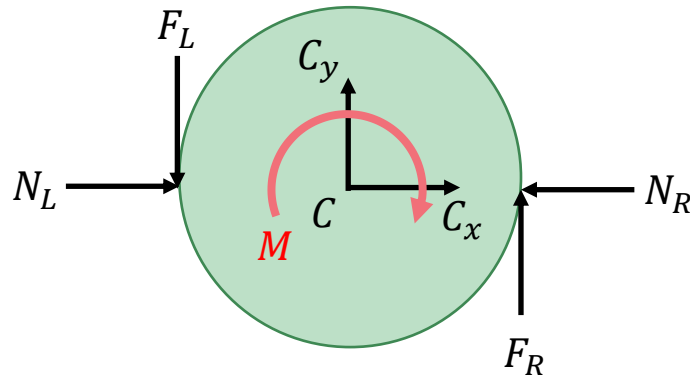
Finally, substituting the two forces into the first equation, gives

$$0.1538F_{HC} + 0.1177F_{HC} = 400 \Rightarrow F_{HC} = 1473 \text{ N}$$



Example 4.9...

(b) Consider the *FBD* of the drum.



Taking moment about C, we have

$$(0.25)(F_L + F_R) - M = 0$$

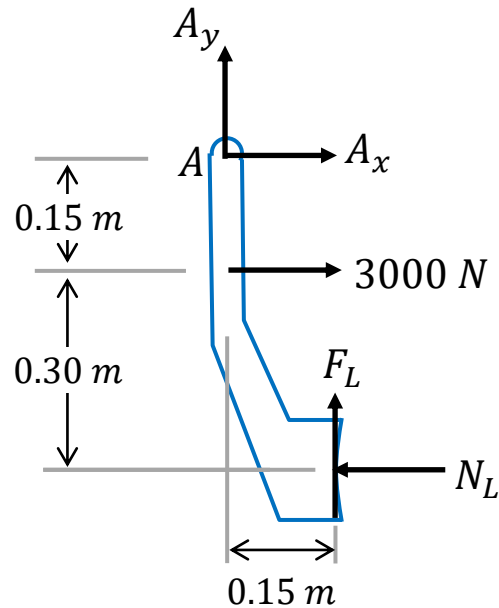
$$\Rightarrow M = 0.25(F_L + F_R)$$

Now, the drum is rotating. Hence, this implies

$$F_L = \mu_k N_L = 0.3N_L \Rightarrow N_L = 0.3333F_L$$

$$F_R = 0.4N_R \Rightarrow N_R = 0.3333F_R$$

Example 4.9



Next, consider the *FBD* of the left arm. For the moment about A, gives

$$\begin{aligned} (3000)(0.15) + F_L(0.15) - N_L(0.45) &= 0 \\ \Rightarrow (3000)(0.15) + F_L(0.15) - (3.3333F_L)(0.45) &= 0 \\ \Rightarrow F_L &= 333.3 \text{ N} \end{aligned}$$

Similarly for the right arm, we have

$$\begin{aligned} (3000)(0.15) - F_R(0.15) - N_R(0.45) &= 0 \\ \Rightarrow F_R &= 272.7 \text{ N} \end{aligned}$$

Finally, substituting the two forces into the moment, gives

$$M = 0.25(333.3 + 272.7) = 151.5 \text{ N.m}$$

